

Cherry Blossom – A formula for SE in difference of proportions

MATH 213

As you saw in the last activity, the sampling distribution for the difference in proportions seems to follow the same "central-limit theorem" pattern as the sampling distribution for a sample mean or a sample proportion:

- Shape is roughly normal.
- Sampling distribution is centered at the population parameter.

To use our standard procedure for creating confidence intervals, we need a formula for the standard error in the sampling distribution. Here is the way that formula works:

- First calculate the standard error for a sample proportion using samples of size 50 from the 10-mile group. Call this standard error SE_1.
- Then calculate the standard error for a sample proportion using samples of size 50 from the 5K group. Call this standard error SE_2.
- The standard error for the difference in proportions between the two samples is

We can think of that formula for stan

$$SE_{\text{diff}} = \sqrt{SE_1^2 + SE_2^2}$$

in Theorem" for standard errors.

Group 1

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?

How many of your intervals actually contained the parameter?

Group 2

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.

Put your graph testing the new confidence-interval procedure here.

How many of the 100 95% intervals do you expect to contain the population parameter?

How many of your intervals actually contained the parameter?

Group 3

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

$$\begin{aligned} \text{SE}_{10\text{mile}} &= \text{square root}(P_{10\text{mile}} (1-P_{10\text{mile}})/n) \\ \text{Square root}(0.599 * (1-0.599)/50) &= 0.0693 \end{aligned}$$

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

$$\begin{aligned} \text{SE}_{5\text{mile}} &= \text{square root}(P_{5\text{mile}} (1-P_{5\text{mile}})/n) \\ \text{SE}_{5k} &= \text{square root}(0.723*(1-0.723)/50) = 0.0632 \end{aligned}$$

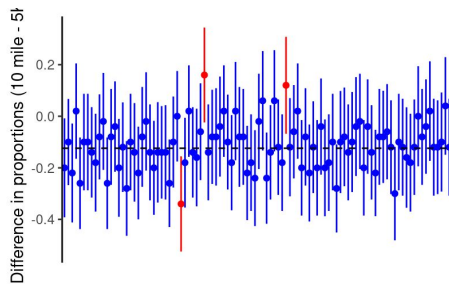
What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

$$\begin{aligned} \text{SE}_{\text{diff}} &= \text{square root}(\text{SE}_{10\text{mile}}^2 + \text{SE}_{5k}^2) \\ \text{SE}_{\text{diff}} &= \text{square root}(0.0693^2 + (0.0632)^2) = 0.0938 \end{aligned}$$

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
 z^*

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.
Using variable `se_diff`

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?
95 out of 100

How many of your intervals actually contained the parameter?
97 out of 100

Group 4

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

$$SE_{10} = 0.069$$

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

$$SE_5 = 0.063$$

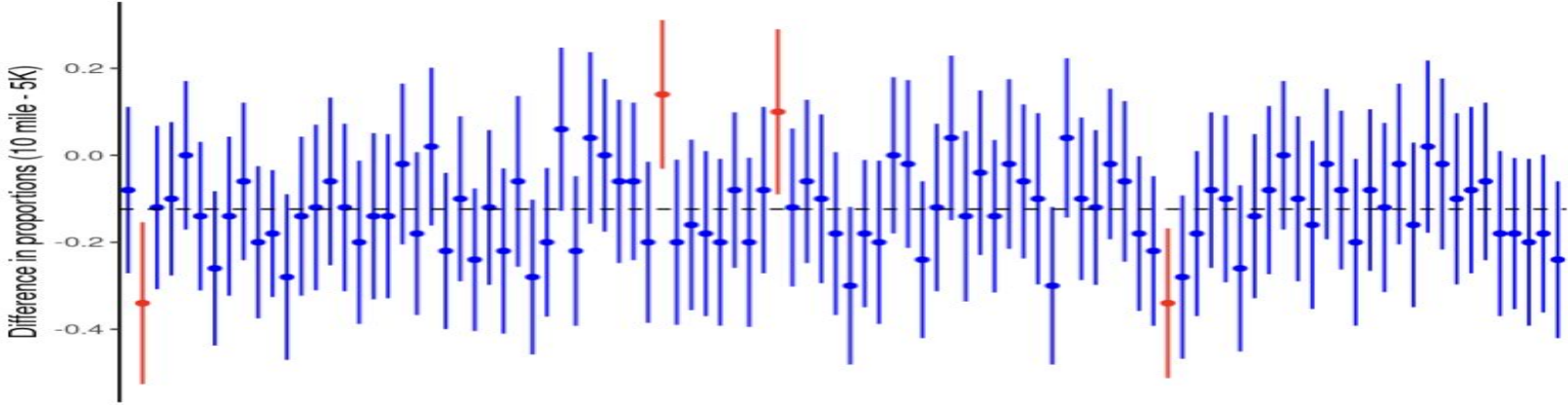
What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

$$SE_{(10-5)} = ((0.069)^2 * (0.063)^2)^{0.5} = 0.0934$$

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
 z^* because 2

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.
SE_diff, was calculated when merge_sum created

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter? **Roughly 95**

How many of your intervals actually contained the parameter? **96**

Group 5

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

SE = 0.069

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

SE = 0.063

What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

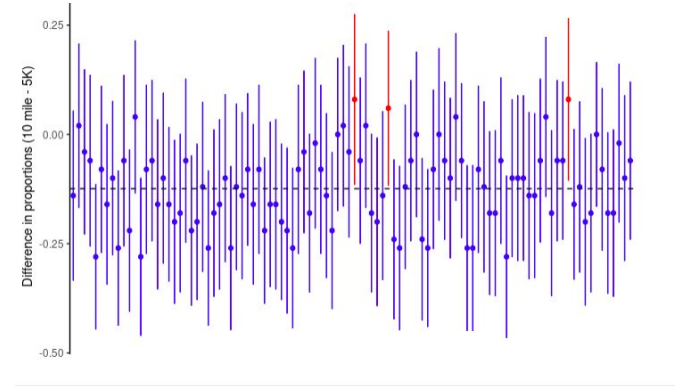
$\text{sqrt}((0.069)^2 + (0.063)^2) = 0.093$

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
Its using z^* since its for a normal distribution

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.

```
se_diff = sqrt(se_10Mile^2 + se_5K^2))
```

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?

95

How many of your intervals actually contained the parameter?

97

Group 6

Try out the formula for the standard error for a difference in proportions:

$$SE_{\text{diff}} = \sqrt{SE_1^2 + SE_2^2}$$

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

0.069

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

0.063

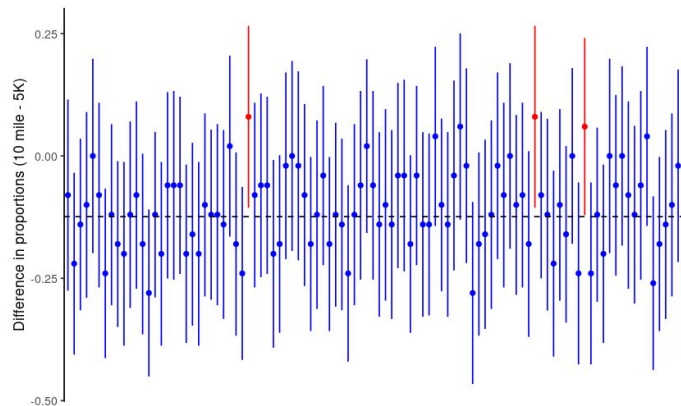
What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

$\text{sqrt}((0.069)^2 + (0.063)^2)$
= 0.093

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
 z^* because it's for a normal distribution

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.
 $\text{sqrt}(\text{se_10Mile}^2 + \text{se_5K}^2)$

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?
95

How many of your intervals actually contained the parameter? 97

Group 7

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

0.069

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

0.063

What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

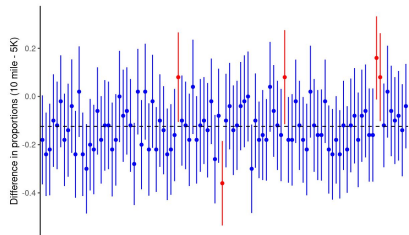
$$Se = \sqrt{(0.069)^2 + (0.063)^2}$$

Se = 0.093

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
z

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.
0.0958

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?
95

How many of your intervals actually contained the parameter?
95

Group 8

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

.069

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

.063

What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

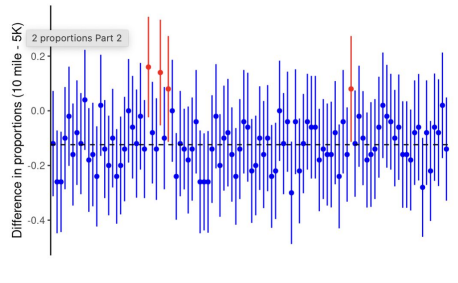
.006

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?

z value

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top. The se we used was `se_diff`, and it was calculated by squaring the se of each 5k and 10k then subtracting 5k from 10k then taking the square root

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?

About 95

How many of your intervals actually contained the parameter?

96

Group 9

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.)

0.069

What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.)

0.063

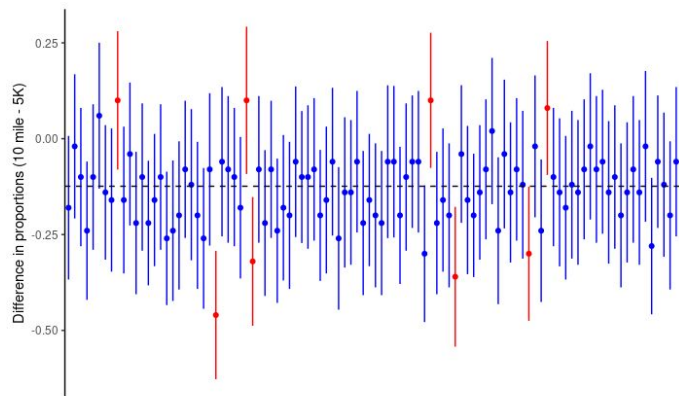
What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!)

$$\sqrt{(0.069^2 + 0.063^2)} = \sqrt{(0.004761 + 0.003969)} = \sqrt{(0.00873)} = 0.09343446901$$

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
Neither? It uses 2

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.
The standard error of the difference: $\sqrt{SE_{10}^2 + SE_5^2}$

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?

95

How many of your intervals actually contained the parameter?

92

Group 10

Try out the formula for the standard error for a difference in proportions:

What is the standard error for the sample proportion of “F” in samples of size 50 from the 10-mile group? (We did this in an earlier activity.) .069

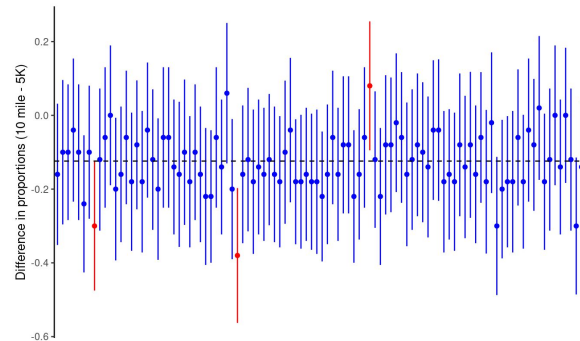
What is the standard error for the sample proportion of “F” in samples of size 50 from the 5K group? (We did this in an earlier activity.) .063

What is the standard error for the difference in proportions of “F” for samples of size 50 from the 10-mile group and the 5K group? (Show some of your work!) $\sqrt{.069^2 + .063^2} = 0.09343446901$

Look at the code to generate 100 confidence intervals. Is it using a z^* value or a t^* value as a coefficient?
Neither I don't think. I think it's using 2. But technically it is z distribution

What standard error is it using? Where was that calculated? Hint: you'll need to scroll back up to the top.
`Se_diff`. It is found in the merged sum

Put your graph testing the new confidence-interval procedure here.



How many of the 100 95% intervals do you expect to contain the population parameter?
About 95

How many of your intervals actually contained the parameter? 97